Cove task cards

Nov 9, 2026 · @Qadeer

Every task in the one-month plan as a kanban card: owner, week, a rough estimate, what it depends on, the steps, and a done-when check you can test.

How to use these cards

Copy each card below onto your board as one card: the heading is the card title, and the steps become its checklist.

Columns: Backlog → This week → Doing → Review → Done.

Labels:

- Owner: You, Teammate 2, Teammate 3, All
- Week: W1, W2, W3, W4
- Area: Kickoff, Core, UI, Voice, Files, System, Ship
- Stretch: only for cards to start once everything else that week is in Review or Done

Rules:

- **At most 2 cards in Doing per person.** Finish before starting.
- **A card moves to Review** when its pull request is open. It moves to Done when it's merged and its done-when check passes.
- **Each Sunday,** move the next week's cards from Backlog to This week.
- **Don't start a card** until the cards it depends on are in Done, or a fake from K-11 stands in for them.
- **Card IDs:** K = kickoff, C = core and UI, V = voice, F = file engine, S = system modules, Q = shipping.

Hours per person

The estimates add up to about 25 to 30 hours a week each, with your load the highest in weeks 1 and 2. Estimates are rough guesses to balance the work, not deadlines.

Person	Week 1	Week 2	Week 3	Week 4	Total	Stretch
You	35 h	30 h	29 h	13 h	107 h	+10 h

Person	Week 1	Week 2	Week 3	Week 4	Total	Stretch
Teammate 2	30 h	23 h	29 h	11 h	93 h	+14 h
Teammate 3	24 h	21 h	31 h	21 h	97 h	None

Week totals include kickoff cards in week 1 and the Sunday demo every week. If you fall behind in week 1, Teammate 3 has the most room to take a card from your lane.

All cards at a glance

Eighty cards in total, sorted by week and then by owner. Stretch cards are marked; the full details for each card follow below.

ID	Card	Owner	Week	Est. (h)	Depends on
K-01	Set up the GitHub repo and board	All	Kickoff	1	None
K-02	Create the .NET solution	Teammate 2	Kickoff	2	K-01
K-03	Automatic build and tests	Teammate 2	Kickoff	2	K-02
K-04	Code style and pull request template	Teammate 3	Kickoff	1	K-01
K-05	Everyone runs the empty app	All	Kickoff	1	K-02
K-06	Draft the core contracts	You	Kickoff	4	K-02
K-07	Draft the job and file contracts	Teammate 2	Kickoff	2	K-02
K-08	Approve the contracts	All	Kickoff	1	K-06, K-07
K-09	Sketch the pill and Files panel in Figma	You	Kickoff	3	None
K-10	Register the OneDrive app	Teammate 2	Kickoff	1	None
K-11	Build fakes for parallel work	Teammate 3	Kickoff	2	K-08
C-01	Tray app and single instance	You	1	3	K-08
C-02	Settings store	You	1	3	K-08
C-03	Module loader	You	1	3	C-02
C-04	Event bus	You	1	3	K-08
C-05	Pill window shell	You	1	5	C-01
C-06	Pill states and focus handling	You	1	5	C-05
C-07	Simple alert queue	You	1	2	C-04, C-06
F-01	Job queue	Teammate 2	1	6	K-08
F-02	Safe output and cleanup	Teammate 2	1	2	F-01

ID	Card	Owner	Week	Est. (h)	Depends on
F-03	Image convert	Teammate 2	1	6	F-01
F-04	Command-line test harness	Teammate 2	1	2	F-03
F-05	Image converter tests	Teammate 2	1	3	F-03
S-01	Media session reader	Teammate 3	1	5	K-08
S-02	Quiet-time detector	Teammate 3	1	3	K-08
S-03	Download and screenshot watcher	Teammate 3	1	5	K-08
S-04	Global shortcuts	Teammate 3	1	4	K-08
C-08	UI kit	You	2	8	K-09
C-09	Files panel layout	You	2	8	C-08
C-10	Files panel wiring	You	2	5	C-09, F-01, F-06
C-11	Alert rules	You	2	4	C-07
C-12	Cove key Menu	You	2	4	C-06, S-04
F-06	Explorer selection	Teammate 2	2	6	K-08
F-07	Copied files	Teammate 2	2	3	K-08
F-08	Dropped files and folders	Teammate 2	2	2	K-08
F-09	Video and audio convert	Teammate 2	2	8	F-01
F-10	Friendly failure messages	Teammate 2	2	3	F-01
S-05	Now Playing panel	Teammate 3	2	6	S-01, C-08
S-06	Song-change alerts	Teammate 3	2	1	S-01, C-07
S-07	Download and screenshot alerts	Teammate 3	2	4	S-03, C-10
S-08	Settings window and Modules page	Teammate 3	2	5	C-03, C-08
S-09	Shortcuts page	Teammate 3	2	4	S-04, S-08
C-13	Peek state	You	3	4	C-06
C-14	Command bar	You	3	6	C-12
C-15	Motion polish and reduced motion	You	3	4	C-06
V-01	Push-to-talk key	You	3	3	C-14
V-02	Speech commands	You	3	6	V-01
V-03	Listening state in the pill	You	3	3	V-01
V-04	Voice tests and tuning	You	3	2	V-02
F-11	Compress video	Teammate 2	3	8	F-09

ID	Card	Owner	Week	Est. (h)	Depends on
F-12	Compress images	Teammate 2	3	3	F-03
F-13	Copy result to the clipboard	Teammate 2	3	1	F-11
F-14	Copy text with OCR	Teammate 2	3	4	F-01
F-15	OneDrive sign-in	Teammate 2	3	4	K-10
F-16	OneDrive upload and share link	Teammate 2	3	6	F-15
F-17	File commands for the command bar and voice	Teammate 2	3	2	C-14
S-10	Mixer	Teammate 3	3	6	C-08
S-11	Default device switching	Teammate 3	3	3	S-10
S-12	Mic indicator	Teammate 3	3	4	C-13
S-13	Global mic mute	Teammate 3	3	2	S-04, S-12
S-14	Timers and Pomodoro	Teammate 3	3	6	C-13
S-15	Quiet-time rules in the alert queue	Teammate 3	3	3	S-02, C-11
S-16	Settings pages for every module	Teammate 3	3	6	S-08
C-16	Pinning (stretch)	You	4	2	C-06
C-17	Keyboard and screen-reader pass	You	4	4	All UI cards
V-05	"Hey Cove" wake word (stretch)	You	4	8	V-02
V-06	Voice settings page	You	4	2	V-02, S-16
F-18	Shelf (stretch)	Teammate 2	4	6	C-09
F-19	Documents to PDF (stretch)	Teammate 2	4	3	F-01
F-20	Right-click menu (stretch)	Teammate 2	4	5	C-10
F-21	Format tests and free-space check	Teammate 2	4	4	F-11
F-22	Measure performance	Teammate 2	4	3	All W3 cards
S-17	Installer and start with Windows	Teammate 3	4	6	All W3 cards
S-18	First-run tour	Teammate 3	4	4	C-06
S-19	Clean-PC test	Teammate 3	4	3	S-17
Q-01	Sunday demo (weekly)	All	1 to 4	1 each	That week's cards
Q-02	Feature freeze and bug bash	All	4	3 each	All v1.0 cards
Q-03	README	Teammate 3 lead	4	3	Q-02

ID	Card	Owner	Week	Est. (h)	Depends on
Q-04	Demo video	You	4	3	Q-02
Q-05	Publish the v1.0 release	Teammate 3	4	1	S-19, Q-02

Kickoff cards (Nov 9 and 10)

These eleven cards get the project running and the contracts agreed, so everyone can work in parallel from day 3.

K-01 · Set up the GitHub repo and board

Owner: All · **Week:** Kickoff · **Estimate:** 1 h · **Depends on:** none

- [ ] Create the repo "cove" and add all three as collaborators
- [ ] Protect main: pull request required, 1 approval, build must pass
- [ ] Create the board with the columns and labels from How to use these cards
- [ ] Add a .gitignore for Visual Studio and .NET

Done when: a direct push to main is refused, and the board shows its five columns.

K-02 · Create the .NET solution

Owner: Teammate 2 · **Week:** Kickoff · **Estimate:** 2 h · **Depends on:** K-01

- [ ] Create Cove.sln targeting .NET 10, Windows 10 version 19041 or later
- [ ] Projects: Cove.App (WPF startup), Cove.Core, Cove.UI, Cove.Files, Cove.System, Cove.Voice (class libraries), Cove.Tests (xUnit)
- [ ] Add references: App uses all; every module uses Core and UI only
- [ ] Add the NuGet packages from the Tech stack table

Done when: the solution builds with no warnings on all three PCs.

K-03 · Automatic build and tests

Owner: Teammate 2 · **Week:** Kickoff · **Estimate:** 2 h · **Depends on:** K-02

- [ ] GitHub Actions workflow on windows-latest: restore, build, run tests
- [ ] Runs on every pull request and every push to main
- [ ] Make it a required check in the branch rules

Done when: a pull request with a failing test cannot be merged.

K-04 · Code style and pull request template

Owner: Teammate 3 · **Week:** Kickoff · **Estimate:** 1 h · **Depends on:** K-01

- [ ] .editorconfig with the standard C# naming rules
- [ ] Pull request template: what changed, card ID, how to test, screenshot for UI
- [ ] Branch naming in the README: feature/<area>-<thing>, fix/<area>-<thing>
- [ ] docs/decisions.md with its first entry: "C#, .NET 10, WPF"

Done when: a new pull request opens with the template filled in.

K-05 · Everyone runs the empty app

Owner: All · **Week:** Kickoff · **Estimate:** 1 h · **Depends on:** K-02

- [ ] Clone, build and run on each PC
- [ ] Note any setup problem in the README's "Getting started"

Done when: all three see the app start and quit cleanly.

K-06 · Draft the core contracts

Owner: You · **Week:** Kickoff · **Estimate:** 4 h · **Depends on:** K-02

- [ ] IModule: Id, DisplayName, Enable(), Disable(), settings type
- [ ] CoveEvent: Type, Source, Priority, Data, up to two actions (label plus callback)
- [ ] IEventBus: Publish(event), Subscribe(type, handler)
- [ ] IPanel: Title, Menu letter, CreateView()
- [ ] ICommand: Name, aliases, arguments, Run(args)
- [ ] ISettingsStore: Get<T>(module), Save(module, value)
- [ ] One short doc comment on each, with an example

Done when: the contracts compile in Cove.Core and are in a pull request for K-08.

K-07 · Draft the job and file contracts

Owner: Teammate 2 · **Week:** Kickoff · **Estimate:** 2 h · **Depends on:** K-02

- [ ] IJobService: Start(job) returns an id; Progress, Completed and Failed events; Cancel(id)

- [ ] Job: kind (convert, compress, upload, text), input files, options
- [ ] JobResult: output paths, link, copied text, error message
- [ ] IFileSource: GetCurrentFiles() returning Explorer selection, recently copied files, or nothing

Done when: the contracts compile and are in the same review as K-06.

K-08 · Approve the contracts

Owner: All · **Week:** Kickoff · **Estimate:** 1 h · **Depends on:** K-06, K-07

- [ ] Walk through each interface together on a call
- [ ] Each person confirms they can build their week 1 cards against them
- [ ] Merge, and record "contracts approved" in docs/decisions.md

Done when: the contracts are merged with all three approvals.

K-09 · Sketch the pill and Files panel in Figma

Owner: You · **Week:** Kickoff · **Estimate:** 3 h · **Depends on:** none

- [ ] Frames for Hidden, Peek, Alert, Menu and Panel, at the spec's sizes
- [ ] The Files panel: header, file cards, action row, options, Run
- [ ] A small style sheet: colours, corner radius, text sizes, spacing
- [ ] Share the link in the README

Done when: teammates can open the file and build their panels in the same style.

K-10 · Register the OneDrive app

Owner: Teammate 2 · **Week:** Kickoff · **Estimate:** 1 h · **Depends on:** none

- [ ] Register "Cove" in the Microsoft Entra admin center (free with a Microsoft account)
- [ ] Allow personal Microsoft accounts; add a desktop redirect URI
- [ ] Add the Files.ReadWrite permission
- [ ] Keep the client ID out of the repo (local config file, git-ignored)

Done when: a test sign-in in a console app returns a token. Doing this on day 2 avoids a week-3 surprise.

K-11 · Build fakes for parallel work

Owner: Teammate 3 · **Week:** Kickoff · **Estimate:** 2 h · **Depends on:** K-08

- [ ] FakeJobService: pretends to convert, with progress over 5 s, then succeeds or fails on request
- [ ] FakeFileSource: returns two sample files
- [ ] FakeEventSource: publishes a song change and a download every 10 s
- [ ] A setting that switches fakes on for development builds

Done when: the pill can show a fake alert and a fake job's progress before any real module exists.

Your cards: core and UI

Seventeen cards that build the app's skeleton and everything people see. Week 1 is the foundation, so keep each card small and merge it the day it works.

C-01 · Tray app and single instance

Owner: You · **Week:** 1 · **Estimate:** 3 h · **Depends on:** K-08

- [ ] Tray icon with H.NotifyIcon; right-click menu with Settings and Quit
- [ ] Only one copy can run (a named mutex); a second launch tells the first to open the Menu
- [ ] No taskbar button and no main window on start
- [ ] Quit stops every module cleanly

Done when: launching Cove twice leaves one tray icon, and Quit leaves no process behind.

C-02 · Settings store

Owner: You · **Week:** 1 · **Estimate:** 3 h · **Depends on:** K-08

- [ ] Implement ISettingsStore with System.Text.Json, saved to AppData\Cove\settings.json
- [ ] One section per module, with defaults when missing
- [ ] Save a few seconds after a change, not on every keystroke
- [ ] A broken file is backed up and replaced by defaults, never a crash

Done when: a setting survives a restart, and deleting half the file still lets Cove start.

C-03 · Module loader

Owner: You · **Week:** 1 · **Estimate:** 3 h · **Depends on:** C-02

- [ ] Register every IModule at startup
- [ ] Enable or disable each one from its saved on/off setting
- [ ] A module that throws is disabled and logged with Serilog; the rest keep running

Done when: a test module that throws on Enable leaves Cove running and writes a clear log line.

C-04 · Event bus

Owner: You · **Week:** 1 · **Estimate:** 3 h · **Depends on:** K-08

- [ ] Implement IEventBus: thread-safe publish and subscribe by event type
- [ ] Handlers that touch the UI run on the UI thread
- [ ] A failing handler is logged and doesn't stop the others
- [ ] Unit tests for ordering and failures

Done when: events published from a background thread reach a UI handler without errors.

C-05 · Pill window shell

Owner: You · **Week:** 1 · **Estimate:** 5 h · **Depends on:** C-01

- [ ] Borderless, transparent WPF window with fully rounded ends
- [ ] Always on top, no taskbar button, hidden from Alt+Tab
- [ ] Never takes focus when shown (no-activate window style through CsWin32)
- [ ] Centred at the top of the main monitor, 8 px from the edge, correct at 100% to 200% scaling

Done when: the pill appears over a maximised Chrome window, and typing in Notepad continues uninterrupted.

C-06 · Pill states and focus handling

Owner: You · **Week:** 1 · **Estimate:** 5 h · **Depends on:** C-05

- [ ] States Hidden, Alert and Panel, with a simple slide and resize between them

- [ ] Panel takes keyboard focus; closing it gives focus back to the window that had it
- [ ] Esc and a click outside close the panel
- [ ] Host any IPanel's view inside the Panel state

Done when: opening and closing a test panel from Word returns the cursor to the same spot in Word.

C-07 · Simple alert queue

Owner: You · **Week:** 1 · **Estimate:** 2 h · **Depends on:** C-04, C-06

- [ ] Subscribe to alert events; show one at a time in the Alert state
- [ ] Icon, one-line message, up to two buttons; auto-hide after 4 s
- [ ] Buttons run their callbacks

Done when: three fake alerts in a row show one after another, and a button click runs its action.

C-08 · UI kit

Owner: You · **Week:** 2 · **Estimate:** 8 h · **Depends on:** K-09

- [ ] Theme from the Figma style sheet, following Windows light or dark and the accent colour (WPF-UI)
- [ ] Controls: button, icon button, file card, progress bar, list row, toggle, slider, text field
- [ ] Panel template: title row, content area, footer with the main button
- [ ] A hidden "kit" panel showing every control, for teammates to copy from

Done when: Teammate 3 builds the Now Playing panel with only kit controls.

C-09 · Files panel layout

Owner: You · **Week:** 2 · **Estimate:** 8 h · **Depends on:** C-08

- [ ] Header with file count and a pin button (inactive until C-16)
- [ ] File cards with thumbnail, name, type, size, folder, remove button
- [ ] Drop area that highlights while a file is dragged over it
- [ ] Action row: 1 Convert, 2 Compress, 3 Upload, 4 Copy text, 5 Shelf (greyed out until F-18), 6 More
- [ ] Options area that changes per action: formats for Convert, presets for Compress
- [ ] Keyboard: number keys, arrows, Enter, Delete, Esc

Done when: the panel works fully with the keyboard on fake files.

C-10 · Files panel wiring

Owner: You · **Week:** 2 · **Estimate:** 5 h · **Depends on:** C-09, F-01, F-06

- [ ] On open, load files from IFileSource; show the "Use 2 copied files" chip when offered
- [ ] Pass dropped files to Teammate 2's drop handler (F-08)
- [ ] Run starts a job through IJobService; show progress, done and failed states
- [ ] Cancel button; closing the panel keeps the job running
- [ ] OpenFilesPanel(paths) so alerts can open the panel with files loaded

Done when: select a photo, Ctrl+Alt+D, 1, Enter gives "Saved photo.jpg" with the real engine.

C-11 · Alert rules

Owner: You · **Week:** 2 · **Estimate:** 4 h · **Depends on:** C-07

- [ ] Priority order from the spec's Automatic pop-ups table
- [ ] Merge alerts of the same type within 5 s ("3 files sorted")
- [ ] 4 s normal, 8 s urgent; hovering pauses the countdown
- [ ] Unit tests for priority and merging

Done when: a burst of ten mixed fake alerts shows in priority order with duplicates merged.

C-12 · Cove key Menu

Owner: You · **Week:** 2 · **Estimate:** 4 h · **Depends on:** C-06, S-04

- [ ] Ctrl+Alt+Space opens the Menu state with a letter for each registered IPanel
- [ ] A letter opens that panel; Esc closes; the same shortcut again closes
- [ ] Letters come from each module, so new panels appear without changing the Menu

Done when: Ctrl+Alt+Space then D opens the Files panel, and M opens Now Playing once S-05 lands.

C-13 · Peek state

Owner: You · **Week:** 3 · **Estimate:** 4 h · **Depends on:** C-06

- [ ] A small status line modules can add to: song title, mic dot, job spinner and count, timer countdown
- [ ] Shows while a job or timer runs after the panel closes; hides 3 s after there's nothing to show
- [ ] Lets clicks pass through except on its icons

Done when: closing the Files panel mid-job shows Peek with a spinner until the job finishes.

C-14 · Command bar

Owner: You · **Week:** 3 · **Estimate:** 6 h · **Depends on:** C-12

- [ ] / in the Menu switches to a text field
- [ ] Parse "verb arguments" and find the ICommand by name or alias
- [ ] Suggestions while typing, Tab completes, Enter runs, Up brings back the last 20 commands
- [ ] Unknown command shows "No command called …" with the closest match
- [ ] A command router that voice (V-02) can call with the same text

Done when: "timer 10" and "compress discord" both run from the bar once their modules register them.

C-15 · Motion polish and reduced motion

Owner: You · **Week:** 3 · **Estimate:** 4 h · **Depends on:** C-06

- [ ] Easing and a slight spring on open and close, about 250 ms in and 180 ms out
- [ ] No flicker or jump when the pill changes size
- [ ] Reduced-motion setting, defaulting to Windows' own animation setting: fade only

Done when: a screen recording at 60 fps shows smooth transitions with no flashes.

C-16 · Pinning (stretch)

Owner: You · **Week:** 4 · **Estimate:** 2 h · **Depends on:** C-06

- [ ] Pin button and Ctrl+P keep the panel open when clicking elsewhere
- [ ] Pinned panels don't take focus back after you click away

Done when: a pinned Files panel stays open while you drag files in from Explorer.

C-17 · Keyboard and screen-reader pass

Owner: You · **Week:** 4 · **Estimate:** 4 h · **Depends on:** all UI cards

- [ ] Every panel usable with only the keyboard, with a visible focus outline
- [ ] Every control has a name for screen readers
- [ ] Check high-contrast mode once

Done when: Windows Narrator reads every Files panel control by name.

Your cards: voice

Six cards, all offline. Voice reuses the command bar's router (C-14), so a spoken "compress" runs exactly what typing "compress" runs, and every new command works by voice automatically.

V-01 · Push-to-talk key

Owner: You · **Week:** 3 · **Estimate:** 3 h · **Depends on:** C-14

- [ ] Ctrl+Alt+V starts listening (registered through S-04's shortcut service)
- [ ] Listening stops on silence, after 4 s, or on Esc
- [ ] Optional hold mode: listen only while the keys are held, by checking the key state every 30 ms
- [ ] Only one listening session at a time

Done when: pressing Ctrl+Alt+V in any app starts a listening session, and Esc cancels it.

V-02 · Speech commands

Owner: You · **Week:** 3 · **Estimate:** 6 h · **Depends on:** V-01

- [ ] System.Speech recogniser on the default mic, English, fully offline
- [ ] Build the grammar from every registered ICommand name and alias, so new commands are recognised automatically
- [ ] Allowed words for arguments: formats (jpg, png, pdf, mp3, mp4), presets (discord, email), numbers one to sixty plus "minutes"
- [ ] Turn the heard phrase into command text ("timer ten minutes" becomes "timer 10") and pass it to the router
- [ ] Ignore results below a confidence threshold; start around 0.6 and tune in V-04

Done when: saying "compress", "convert to pdf", "timer ten minutes" and "mute" each run the right command.

V-03 · Listening state in the pill

Owner: You · **Week:** 3 · **Estimate:** 3 h · **Depends on:** V-01

- [ ] The pill shows "Listening" with a live mic-level bar (from the recogniser's audio-level event)
- [ ] Shows what it heard for a moment before running it
- [ ] "Didn't catch that" with a hint of example commands when nothing matches
- [ ] Cove appears in its own mic indicator while listening, so users can trust it

Done when: a teammate who has never used it can tell when Cove is listening and what it heard.

V-04 · Voice tests and tuning

Owner: You · **Week:** 3 · **Estimate:** 2 h · **Depends on:** V-02

- [ ] Unit tests: heard phrase to command text for every command
- [ ] A 20-phrase script; each of the three of you reads it in a quiet room and with music playing
- [ ] Write down how many phrases worked; rename or drop commands that get confused
- [ ] Tune the confidence threshold from the results

Done when: the test script works for all three voices in a quiet room, and the results are noted in docs/decisions.md.

V-05 · "Hey Cove" wake word (stretch)

Owner: You · **Week:** 4 · **Estimate:** 8 h · **Depends on:** V-02

- [ ] Create a "Hey Cove" keyword for Windows in the Picovoice Console; check the free tier's terms fit a student project
- [ ] Run Porcupine on mic audio in the background; on detection, start a V-02 listening session
- [ ] Off by default; when off, the mic is fully released
- [ ] Keep the access key in a local, git-ignored config file
- [ ] Check CPU use while idle and listening

Done when: "Hey Cove, compress" opens Compress hands-free, and turning the setting off removes Cove from the mic indicator.

V-06 · Voice settings page

Owner: You · **Week:** 4 · **Estimate:** 2 h · **Depends on:** V-02, S-16

- [ ] On or off, mode (press to talk, hold to talk, wake word if V-05 ships), the shortcut
- [ ] Confidence slider and a "Test" button that shows what was heard
- [ ] The list of commands you can say, generated from the registered commands

Done when: every voice option can be changed in Settings and survives a restart.

Teammate 2 cards: file engine

Twenty-two cards covering everything that reads, changes or uploads files. Almost all of it can be tested from the command-line harness (F-04) before the UI exists.

F-01 · Job queue

Owner: Teammate 2 · **Week:** 1 · **Estimate:** 6 h · **Depends on:** K-08

- [ ] Implement IJobService: a queue running at most two jobs at once
- [ ] Each job reports progress (0 to 100, or unknown) and can be cancelled
- [ ] Jobs run on background threads; events reach the UI through the event bus
- [ ] Video jobs run at below-normal process priority so the PC stays responsive

Done when: five queued fake jobs run two at a time, report progress, and one can be cancelled midway.

F-02 · Safe output and cleanup

Owner: Teammate 2 · **Week:** 1 · **Estimate:** 2 h · **Depends on:** F-01

- [ ] Write output to a temporary name, then rename it when the job succeeds
- [ ] Delete the temporary file on cancel or failure
- [ ] Output sits next to the original; if the name is taken, add " (1)", " (2)" and so on
- [ ] Never overwrite any existing file

Done when: converting the same file three times gives three files, and a cancelled job leaves nothing behind.

F-03 · Image convert

Owner: Teammate 2 · **Week:** 1 · **Estimate:** 6 h · **Depends on:** F-01

- [ ] Magick.NET: from PNG, JPG, WebP, HEIC, BMP, GIF, TIFF to PNG, JPG, WebP, ICO, PDF
- [ ] Options: quality (JPG, WebP) and an optional maximum width or height
- [ ] Keep orientation from phone photos (EXIF rotation)
- [ ] Register "convert" as a job kind

Done when: an iPhone HEIC photo converts to an upright JPG.

F-04 · Command-line test harness

Owner: Teammate 2 · **Week:** 1 · **Estimate:** 2 h · **Depends on:** F-03

- [ ] A small console project: cove-cli convert <file> --to jpg, and later compress, upload, text
- [ ] Prints progress and the result path

Done when: every file action can be tried from the terminal without opening Cove.

F-05 · Image converter tests

Owner: Teammate 2 · **Week:** 1 · **Estimate:** 3 h · **Depends on:** F-03

- [ ] A tests/samples folder with one small file per input format
- [ ] xUnit tests: each conversion produces a readable file of the right type
- [ ] Tests for naming and cleanup from F-02

Done when: the image tests pass in GitHub Actions.

F-06 · Explorer selection

Owner: Teammate 2 · **Week:** 2 · **Estimate:** 6 h · **Depends on:** K-08

- [ ] Remember which window was in front when the shortcut was pressed, before Cove's panel takes focus
- [ ] If it's an Explorer window, read its selected files through the Windows Shell (IShellWindows)
- [ ] Handle files selected on the desktop
- [ ] Return nothing, not an error, for any other app

Done when: selecting three files in Explorer and pressing Ctrl+Alt+D loads exactly those three.

F-07 · Copied files

Owner: Teammate 2 · **Week:** 2 · **Estimate:** 3 h · **Depends on:** K-08

- [ ] Listen for clipboard changes (AddClipboardFormatListener on a hidden window)
- [ ] When the clipboard holds files, remember them with the time
- [ ] Offer them through IFileSource for 2 minutes, then forget them

Done when: Ctrl+C on two files, then Ctrl+Alt+D, shows "Use 2 copied files".

F-08 · Dropped files and folders

Owner: Teammate 2 · **Week:** 2 · **Estimate:** 2 h · **Depends on:** K-08

- [ ] Accept dropped files from the Files panel's drop area
- [ ] A dropped folder adds the files directly inside it; subfolders only if the setting is on
- [ ] Skip shortcuts and hidden system files

Done when: dropping a folder of 10 photos adds 10 file cards.

F-09 · Video and audio convert

Owner: Teammate 2 · **Week:** 2 · **Estimate:** 8 h · **Depends on:** F-01

- [ ] Bundle ffmpeg in a tools folder; call it through FFMpegCore
- [ ] Video: MP4, MOV, MKV, AVI, WebM to MP4, WebM, GIF, and MP3 (audio only)
- [ ] Audio: MP3, WAV, M4A, FLAC, OGG to MP3, WAV, M4A, FLAC
- [ ] Real progress from ffmpeg's position against the file's duration
- [ ] GIF options: start, end, frame rate, width, with a colour palette pass for quality

Done when: a 1-minute MOV becomes an MP4 with a smooth progress bar, and a 5-second clip becomes a clean GIF.

F-10 · Friendly failure messages

Owner: Teammate 2 · **Week:** 2 · **Estimate:** 3 h · **Depends on:** F-01

- [ ] Map errors to plain messages: file open in another app, unsupported or damaged file, not enough disk space, no internet

- [ ] Name the app holding a locked file when Windows reports it
- [ ] Every failure offers Retry, and Details shows the technical error for bug reports

Done when: converting a .docx that's open in Word says "Essay.docx is open in Word" and Retry works after closing it.

F-11 · Compress video

Owner: Teammate 2 · **Week:** 3 · **Estimate:** 8 h · **Depends on:** F-09

- [ ] Presets: Discord (10 MB), Email (25 MB), custom size
- [ ] Read the duration, then video bitrate = target size ÷ duration − audio bitrate, leaving about 5% headroom
- [ ] Two-pass H.264 encode, so the result lands just under the target
- [ ] If the bitrate is too low for the resolution, step down (1080p → 720p → 480p); pick the floors by eye and write them down
- [ ] If even the lowest step can't fit, warn before starting and show the smallest size possible
- [ ] Output named "name (compressed).mp4"

Done when: a 60 MB, 1-minute clip comes out under 10 MB and still looks watchable.

F-12 · Compress images

Owner: Teammate 2 · **Week:** 3 · **Estimate:** 3 h · **Depends on:** F-03

- [ ] Lower JPG or WebP quality step by step, then the size, until under the target
- [ ] Presets: web image (about 500 KB), Email (25 MB), custom
- [ ] Several images in one job each get the same target

Done when: a 6 MB phone photo comes out under 500 KB.

F-13 · Copy result to the clipboard

Owner: Teammate 2 · **Week:** 3 · **Estimate:** 1 h · **Depends on:** F-11

- [ ] Option on Compress and Convert: put the result file on the clipboard
- [ ] The done alert says "Copied"

Done when: after compressing, Ctrl+V in a Discord message attaches the file.

F-14 · Copy text with OCR

Owner: Teammate 2 · **Week:** 3 · **Estimate:** 4 h · **Depends on:** F-01

- [ ] Windows.Media.Ocr in the user's installed languages
- [ ] Images: load, shrink if above the OCR size limit, read text
- [ ] PDFs: render each page with Windows.Data.Pdf, then read it
- [ ] Copy the text and return it for the panel's preview

Done when: a screenshot of a lecture slide gives clean, pasteable text.

F-15 · OneDrive sign-in

Owner: Teammate 2 · **Week:** 3 · **Estimate:** 4 h · **Depends on:** K-10

- [ ] MSAL sign-in in the system browser with the K-10 app
- [ ] Save the token cache encrypted for the Windows user, so sign-in happens once
- [ ] Sign out clears it
- [ ] Expired sign-in raises the "Sign in to OneDrive again" alert

Done when: sign in once, restart Cove, and uploads work without signing in again.

F-16 · OneDrive upload and share link

Owner: Teammate 2 · **Week:** 3 · **Estimate:** 6 h · **Depends on:** F-15

- [ ] Upload into a "Cove" folder in OneDrive through Microsoft Graph
- [ ] Small files in one request; large files in resumable chunks with progress
- [ ] Create a view-only share link and copy it
- [ ] Retry three times on network errors, then fail with Retry

Done when: a 200 MB video uploads with progress, and its link opens in a private browser window.

F-17 · File commands for the command bar and voice

Owner: Teammate 2 · **Week:** 3 · **Estimate:** 2 h · **Depends on:** C-14

- [ ] Register ICommands: convert <format>, compress <preset or size>, upload, text
- [ ] Each uses IFileSource's current files; with none, it opens the Files panel
- [ ] Aliases that sound clear when spoken: "shrink" for compress, "share" for upload

Done when: typing "convert pdf" with a .docx selected starts the job (once F-19 ships; until then, "convert jpg" on an image).

F-18 · Shelf (stretch)

Owner: Teammate 2 · **Week:** 4 · **Estimate:** 6 h · **Depends on:** C-09

- [ ] Shelf panel (Ctrl+Alt+S) built with the UI kit; Shelf action in the Files panel
- [ ] Stores file paths, not copies; shows Missing if a file moves
- [ ] Drag cards out to any folder or app (copy; Shift moves)
- [ ] Clear one, clear all

Done when: files gathered from three folders can be dragged into one Discord message at once.

F-19 · Documents to PDF (stretch)

Owner: Teammate 2 · **Week:** 4 · **Estimate:** 3 h · **Depends on:** F-01

- [ ] Find LibreOffice if installed; otherwise show "Install LibreOffice to enable"
- [ ] Run it headless to convert DOCX, PPTX, XLSX and ODT to PDF, with a 60 s timeout

Done when: a .docx essay becomes a PDF that looks the same.

F-20 · Right-click menu (stretch)

Owner: Teammate 2 · **Week:** 4 · **Estimate:** 5 h · **Depends on:** C-10

- [ ] Register a "Cove" submenu for all files in the classic menu (Show more options), per user, no admin needed
- [ ] Entries: Convert, Compress, Upload, Copy text, Open in Cove
- [ ] Each entry starts a tiny helper that sends its file path to the running Cove over a named pipe
- [ ] Cove groups paths that arrive within 500 ms into one job, since Explorer starts one helper per file
- [ ] The installer adds the menu; uninstall removes it

Done when: right-clicking three files and choosing Cove > Compress runs one job in the pill.

F-21 · Format tests and free-space check

Owner: Teammate 2 · **Week:** 4 · **Estimate:** 4 h · **Depends on:** F-11

- [ ] One test for every v1.0 format pair, plus compress and OCR
- [ ] Before each job, check free space on the output drive against an estimate of the result's size

Done when: every test passes in GitHub Actions, and a nearly full drive gives a clear warning instead of a broken file.

F-22 · Measure performance

Owner: Teammate 2 · **Week:** 4 · **Estimate:** 3 h · **Depends on:** all week 3 cards

- [ ] Log the time from a shortcut press to the pill showing
- [ ] Note memory and CPU use after 10 minutes idle, with every module on
- [ ] Compare with the spec's targets; fix the worst problem or file a card for version 1.1

Done when: the numbers are written in docs/decisions.md and the worst problem is fixed or logged.

Teammate 3 cards: system modules

Nineteen cards connecting Cove to Windows: shortcuts, media, audio, the mic, folders, settings and the installer. These are smaller, separate pieces, so this lane can also absorb a card from anyone who falls behind.

S-01 · Media session reader

Owner: Teammate 3 · **Week:** 1 · **Estimate:** 5 h · **Depends on:** K-08

- [ ] Use Windows' media session manager to find the current player
- [ ] Read title, artist, album art and the source app; watch for changes
- [ ] Play, pause, next, previous and seek through the session
- [ ] Publish SongChanged and PlaybackChanged events
- [ ] Register ICommands: play, pause, next, previous

Done when: skipping a song in Spotify or a YouTube tab logs a SongChanged event with the right title.

S-02 · Quiet-time detector

Owner: Teammate 3 · **Week:** 1 · **Estimate:** 3 h · **Depends on:** K-08

- [ ] Ask Windows whether a fullscreen app, a game, a presentation or Do Not Disturb is active (SHQueryUserNotificationState)
- [ ] Re-check whenever the front window changes, plus every few seconds
- [ ] Publish QuietStarted and QuietEnded events

Done when: a fullscreen YouTube video and a fullscreen game both start quiet time, and leaving fullscreen ends it.

S-03 · Download and screenshot watcher

Owner: Teammate 3 · **Week:** 1 · **Estimate:** 5 h · **Depends on:** K-08

- [ ] Watch the real Downloads folder (ask Windows for its location, since people move it) and Pictures\Screenshots
- [ ] Ignore temporary files: .crdownload, .part, .tmp
- [ ] A file counts as arrived when its size stays the same for 1 s and it can be opened
- [ ] Publish FileArrived with the path and whether it's a download or a screenshot

Done when: downloading a 100 MB file in Chrome produces one FileArrived event, only after the download finishes.

S-04 · Global shortcuts

Owner: Teammate 3 · **Week:** 1 · **Estimate:** 4 h · **Depends on:** K-08

- [ ] A shortcut service other modules use: Register(name, default keys, action)
- [ ] Uses RegisterHotKey on a hidden window (through CsWin32)
- [ ] When a combination is taken, publish a conflict event and mark it in settings
- [ ] Re-register when the user changes a shortcut
- [ ] Defaults from the spec: Ctrl+Alt+Space, D, M, T, S, A; Ctrl+Alt+V for voice

Done when: every default shortcut fires from any app, and a deliberately taken one shows a conflict.

S-05 · Now Playing panel

Owner: Teammate 3 · **Week:** 2 · **Estimate:** 6 h · **Depends on:** S-01, C-08

- [ ] Built only from UI kit controls: art, title, artist, app icon, controls, seek bar
- [ ] A switcher row when more than one app is playing
- [ ] Keys: Space play or pause, Left and Right skip
- [ ] Give Peek the current song title (C-13)
- [ ] Register as the M panel in the Cove key Menu

Done when: Ctrl+Alt+M shows the current Spotify song, and every control works.

S-06 · Song-change alerts

Owner: Teammate 3 · **Week:** 2 · **Estimate:** 1 h · **Depends on:** S-01, C-07

- [ ] Alert on a new track only, not on pause or resume
- [ ] "Title, Artist" for 3 s with Pause and Next buttons

Done when: skipping three songs shows three short alerts, and pausing shows none.

S-07 · Download and screenshot alerts

Owner: Teammate 3 · **Week:** 2 · **Estimate:** 4 h · **Depends on:** S-03, C-10

- [ ] Suggested actions per file type, from the spec's download table (HEIC: Convert to JPG; DOCX: Convert to PDF; video over 25 MB: Compress)
- [ ] Buttons open the Files panel with the file loaded and the action picked
- [ ] Screenshot alerts offer Copy text, which runs straight away

Done when: downloading a HEIC photo shows an alert whose Convert button produces a JPG in two clicks.

S-08 · Settings window and Modules page

Owner: Teammate 3 · **Week:** 2 · **Estimate:** 5 h · **Depends on:** C-03, C-08

- [ ] A normal window with a side menu of pages (WPF-UI navigation)
- [ ] Opens from the tray menu and from comma in the Cove key Menu
- [ ] Modules page: one card per module with a description and an on/off switch, wired to the module loader
- [ ] Remembers its size and position

Done when: switching off Now Playing removes its panel and alerts immediately, and it stays off after a restart.

S-09 · Shortcuts page

Owner: Teammate 3 · **Week:** 2 · **Estimate:** 4 h · **Depends on:** S-04, S-08

- [ ] List every shortcut with its current keys
- [ ] Click to record new keys; reset to default; switch off
- [ ] Show the conflict message next to any shortcut Windows refused

Done when: changing the Files shortcut to Ctrl+Alt+F works at once and survives a restart.

S-10 · Mixer

Owner: Teammate 3 · **Week:** 3 · **Estimate:** 6 h · **Depends on:** C-08

- [ ] A Mixer tab in the Media panel, built from the UI kit
- [ ] One row per app playing sound (NAudio audio sessions): icon, name, slider, mute
- [ ] Rows appear and disappear as apps start and stop sound
- [ ] Master volume slider
- [ ] Register ICommands: vol <number>, vol <app> <number>

Done when: turning Chrome down in Cove matches Windows' own Volume Mixer.

S-11 · Default device switching

Owner: Teammate 3 · **Week:** 3 · **Estimate:** 3 h · **Depends on:** S-10

- [ ] List output and input devices with NAudio
- [ ] Setting the default device needs an undocumented Windows interface (the one EarTrumpet uses); stop after 3 hours if it fights back
- [ ] Fallback: a button that opens Windows' Sound settings

Done when: you can switch from speakers to headphones in Cove, or the fallback button is in place and the decision is noted.

S-12 · Mic indicator

Owner: Teammate 3 · **Week:** 3 · **Estimate:** 4 h · **Depends on:** C-13

- [ ] Read Windows' privacy usage records for the microphone (an app is using it while its stop time is zero)
- [ ] Check every second, or when the records change

- [ ] Red dot in Peek while any app uses the mic; hovering lists the apps
- [ ] "Mic on: Discord" alert when an app starts using it

Done when: joining a Discord call shows the dot and alert within 2 s, and leaving removes the dot.

S-13 · Global mic mute

Owner: Teammate 3 · **Week:** 3 · **Estimate:** 2 h · **Depends on:** S-04, S-12

- [ ] Ctrl+Alt+A mutes or unmutes the default input device for every app (NAudio)
- [ ] Alert says "Muted" or "Live"; the dot turns grey with a slash while muted
- [ ] Register ICommands: mute, unmute

Done when: in a Discord call, Ctrl+Alt+A mutes you, and Discord shows no sound from your mic.

S-14 · Timers and Pomodoro

Owner: Teammate 3 · **Week:** 3 · **Estimate:** 6 h · **Depends on:** C-13

- [ ] Timer panel (T): quick buttons for 5, 10, 15, 25 and 45 minutes, plus custom
- [ ] Pomodoro: 25 minutes work and 5 break, four times, then 15; lengths adjustable
- [ ] Up to three timers, each with an optional label; pause, +5 min, stop
- [ ] Countdown in Peek; the done alert shows even during quiet time, with an optional sound
- [ ] Register ICommands: timer <minutes>, pomodoro, stop timer

Done when: "timer 1" counts down in Peek and alerts after exactly one minute, even during a fullscreen video.

S-15 · Quiet-time rules in the alert queue

Owner: Teammate 3 · **Week:** 3 · **Estimate:** 3 h · **Depends on:** S-02, C-11

- [ ] During quiet time, show only mic, timer and failure alerts
- [ ] Drop song changes; hold everything else
- [ ] When quiet time ends, one catch-up alert: "While you were busy: 2 downloads, 1 job done"
- [ ] Unit tests; you review the pull request, since it changes your alert queue

Done when: downloading during a fullscreen video shows nothing until you leave fullscreen, then one summary alert.

S-16 · Settings pages for every module

Owner: Teammate 3 · **Week:** 3 · **Estimate:** 6 h · **Depends on:** S-08

- [ ] General: start with Windows, theme, replay the tour
- [ ] Pill: size, reduced motion
- [ ] Alerts: Alert, Peek only or Off for each event type
- [ ] Files: default formats and quality, output folder, jobs at once
- [ ] Cloud: OneDrive sign in and out (F-15)
- [ ] Media and Mixer, Mic, Timer: each module's options
- [ ] Everything saved through ISettingsStore

Done when: every v1.0 setting can be changed and survives a restart.

S-17 · Installer and start with Windows

Owner: Teammate 3 · **Week:** 4 · **Estimate:** 6 h · **Depends on:** all week 3 cards

- [ ] Inno Setup script; install per user, so no admin prompt
- [ ] Bundle the app and the tools folder with ffmpeg
- [ ] Start menu shortcut; "Start with Windows" checkbox, wired to the General setting
- [ ] Uninstall removes the startup entry and the right-click menu, and asks before deleting settings
- [ ] GitHub Actions builds the installer whenever a version tag is pushed

Done when: a teammate installs from the .exe, Cove starts after a reboot, and uninstall leaves nothing behind.

S-18 · First-run tour

Owner: Teammate 3 · **Week:** 4 · **Estimate:** 4 h · **Depends on:** C-06

- [ ] On first launch: "Hi, I'm Cove. Press Ctrl+Alt+Space anytime."
- [ ] Four steps in the pill: the Cove key, trying the Files panel, alerts, where Settings is; each can be skipped
- [ ] Ends with the shortcut conflict check
- [ ] Replay from Settings > General

Done when: a fresh install shows the tour once, and it can be replayed.

S-19 · Clean-PC test

Owner: Teammate 3 · **Week:** 4 · **Estimate:** 3 h · **Depends on:** S-17

- [ ] Use Windows Sandbox (Windows Pro) or a spare PC or virtual machine with nothing installed
- [ ] Install, then run every v1.0 card's done-when check as a test script
- [ ] Open an issue for each failure, labelled with the owner

Done when: the test script passes on a clean machine, or every failure has an issue.

Shipping cards

Five cards that turn four weeks of work into something you can show recruiters: weekly demos, a bug bash, the README, the video and the release.

Q-01 · Sunday demo (weekly)

Owner: All · **Week:** 1 to 4 (Nov 15, 22, 29, Dec 6) · **Estimate:** 1 h each · **Depends on:** that week's cards

- [ ] Merge everything by Saturday night
- [ ] On one PC, run each card's done-when check from that week
- [ ] Move anything unfinished back to This week, or cut it using the cut order
- [ ] Pull next week's cards into This week

Done when: the week's build runs, and next week's cards are on the board. Make a fresh copy of this card each week.

Q-02 · Feature freeze and bug bash

Owner: All · **Week:** 4 (Thu Dec 3 to Sat Dec 5) · **Estimate:** 3 h each · **Depends on:** all v1.0 cards

- [ ] Thursday: no new features merge after this point; only fixes
- [ ] Each person tests the other two people's modules from the done-when checks
- [ ] File every bug as an issue with steps to reproduce, labelled with the owner
- [ ] Fix in order: crashes, data loss, broken demo steps, then everything else

Done when: no open crash or data-loss bugs remain by Saturday night.

Q-03 · README

Owner: Teammate 3 leads; everyone writes their section · **Week:** 4 · **Estimate:** 3 h · **Depends on:** Q-02

- [ ] One-line pitch and a GIF of the pill in action
- [ ] Features, install steps, the default shortcuts
- [ ] Tech stack and a short architecture note (the diagram from the spec)
- [ ] Team: who built what, one section per person in their own words
- [ ] Known limitations and what's planned for version 1.1

Done when: someone outside the team can install and try Cove from the README alone.

Q-04 · Demo video

Owner: You · **Week:** 4 · **Estimate:** 3 h · **Depends on:** Q-02

- [ ] 60 to 90 seconds, no dead time
- [ ] Show in order: Ctrl+Alt+D converting a HEIC, compressing a video for Discord, a song-change alert, muting the mic, "compress" by voice
- [ ] Record at 1080p; add short captions instead of narration if easier
- [ ] Upload unlisted to YouTube; link it at the top of the README

Done when: the video is linked in the README and each teammate has watched it.

Q-05 · Publish the v1.0 release

Owner: Teammate 3 · **Week:** 4 (Sun Dec 6) · **Estimate:** 1 h · **Depends on:** S-19, Q-02

- [ ] Tag v1.0.0; let GitHub Actions build the installer
- [ ] GitHub Release with the installer, release notes and the demo video link
- [ ] Pin the repo on each person's GitHub profile

Done when: the release page has a working installer that a friend can download and run.